

Network monitoring

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Abstract: Networks

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Introduction

Here, we used simulations to test the efficiency of monitoring species interactions and species richness within BONs, testing several methods for design and optimization.

- Context

- ▶ Biodiversity monitoring should encompass all facets of biodiversity
- ▶ Species interactions are rarely considered in comparison to species richness metrics, in part due to their inherent sampling difficulty
- ▶ Yet, interactions represent an biodiversity component which should be considered when implementing monitoring networks
- ▶ As interactions are inherently challenging to sample, developing methods for optimizing strategies is even more relevant.

- Key questions

- ▶ How should we implement monitoring programs *across space* to efficiently capture network composition and changes as essential components of biodiversity?
- ▶ Can we monitor species interactions as efficiently as species ranges? What would be realistic expectations for monitoring species interactions?

Methods

We explored network sampling using simulations to generate networks, species ranges, and biodiversity observations networks (BONs). We used `SpeciesInteractionSamplers.jl` to generate networks, species ranges, and interaction realization and detection, following the approach introduced by Catchen et al. (2023).

Evaluating metaweb sampling

Our first idea was to investigate the efficiency of sampling networks across space to document a metaweb of interactions. Additionally, we aimed to compare this efficiency with the efficiency at which we can document species richness.

Our first step was to simulate networks with variation across space. To do so, we generated autocorrelated ranges (100 x 100 pixels) representing the distribution of 75 species (an arbitrary number) and generated a metaweb using the niche model with a connectance of 0.2 to define the feasible interactions between the species. Combining the species ranges and the metaweb information, we obtained the list of **possible interactions** in our landscape and

for every pixel in the landscape, based only on the co-occurrence of species with feasible interactions. Next, we incorporated a consideration for species abundances, which influence both interaction realization and detection. For example, rare and low-abundance species are less likely to interact in a location because they are less likely to encounter one another, especially compared to abundant species, potentially leading to neutrally forbidden links (Canard et al. 2012; Catchen et al. 2023). We set an abundance profile for our set of species following a Normalized Log Normal distribution, then drew **realized interactions** at each location given the abundance of species. Finally, we defined the set of **detected interactions** by conditioning the detection of realized interactions on the abundance distribution (even if an interaction is realized in a location, we need to be there at that moment to detect, which is again less likely for low probability events; Catchen et al. (2023)).

Next, we generated spatially balanced Biodiversity Observation Networks (BONs) across our landscape, defining sites where to sample species and interactions. We tested all number of sites between 1 and 100 and used the `BalancedAcceptance` method to ensure spatial balance, aiming to distribute sampling evenly across space (for a given number of sampling sites). We then evaluated the sampling efficiency of each BON in terms of overall species richness and compared to the generated metaweb for a each interaction type. For species richness, we assumed we sampled all species present in a sampled location, combined this information across all sampled sites to determine the number of species encountered through sampling, and compared the result with the total species richness in the landscape (75 species). For interactions, we aggregated each interaction type separately (possible, realized and detected) for a given BON and evaluated the proportion of the metaweb recovered through sampling (and therefore for a given number of site). Since some interactions might never be observed in the simulated landscapes (e.g. interacting species who never co-occur), we also measured the proportion of sampled interactions compared to the number of interactions actually present in the landscape (instead of the metaweb) for each interaction type.

We ran simulations with 100 fully-independent replicates (different metaweb, species ranges, abundance distribution, realized interactions) for each number of site in a BON and for the two interaction references, yielding 20,000 simulations in total. Prior exploratory simulations showed the number of sites in the BON to be the main element affecting the proportion of sampled species or interactions compared to other model parameters such as metaweb connectance and landscape autocorrelation.

63 *Focal sampling on single-species & optimization*

64 In complement to the metaweb sampling approach, we also investigated the sampling effi-
65 ciency on a single-species basis. We simulated a focal sampling where an observer would
66 target one species and aim to retrieve all of its interactions (here known from the metaweb),
67 but might have different information to optimize its sampling. Observation network design
68 and efficiency needs to be evaluated beyond spatial balance, and also regarding their efficiency
69 at meeting biodiversity monitoring objectives (Norman and Poisot 2025). Thus, the idea of our
70 focal sampling exploration is to represent a simplified case with a clear objective (retrieving a
71 species' interactions) and where we do have prior knowledge about the system, either through
72 empirical data or model predictions, which can be used to optimize sampling.

73 Optimizing focal sampling relies on two elements: the sampler algorithm used and the
74 information provided for optimization. Species range represents an attainable level of infor-
75 mation an interested observer could have, for instance based on expert knowledge or species
76 distribution models. Therefore, we assumed the observer knew the species range across
77 the landscape (presence or absence). We tested three sampling algorithms for comparison:
78 simple random sampling (to represent uninformed sampling), weighted balanced acceptance
79 sampling (aiming to balance sites across space while weighting favorably for species ranges)
80 and uncertainty sampling (which targets locations with a higher value in the layer provided).
81 Next, we tested four information layers for optimization: one with the focal species range
82 (our attainable information knowledge for the observer), one with the probabilistic range
83 (for a less informed case, similar to the output of a species distribution model), one with the
84 location of realized interactions for the focal species (to represent a best-case scenario) and
85 one with species richness across the landscape (to represent a holistic level of information for
86 the system).

87 Using the approach described in the previous section, we generated a single* set of metaweb,
88 species ranges, and locations for realized interactions (*counter-verified with 10 separate sets,
89 which yielded conceptually similar results). We selected the species with the highest degree
90 (number of interactions) in the metaweb as the focal species for the simulation. We then
91 generated BONs of 1 to 500 sites (at interval increments of 5 sites) using the three possible
92 algorithms and optimization layers, with 100 replicates for every number of sites. For each
93 BON generated, we extracted the number of unique realized interactions sampled across all
94 sites (essentially the monitored degree for the focal species)

Next, we aimed to verify whether our results were constant when replicating simulations with different simulated metawebs and species ranges. We ran 100 independent simulations, each with 50 replicates for every number of sites (once again from 1 to 500 sites at interval increments of 5 sites). However, doing so raised the issue of how to describe and compare the efficiency of the simulated monitoring across samplers, optimization layers or focal species degree, given how the generated metawebs and landscapes might vary between independent simulations, and especially how to describe the relationship between the number of monitoring sites and the proportion of monitored interactions.

Since the exact proportion of monitored interactions is known in our simulations, we used the following single-parameter equation which generates similar efficiency curves:

$$p = \frac{x}{a + x} \quad (1)$$

where p is the proportion of monitored interactions, x is the number of sites in the BON and a is the efficiency parameter we use for comparison. Here, a is the single parameter describing the shape of the curve, and can be compared across samplers, optimization layers or focal species degree. The lower this parameter, the faster we reach a high proportion of monitored interactions.

For each independent simulation, sampler and optimization layer option, we performed a grid search across possible values of a to identify the curve yielding the best fit with the simulation's result. Upon exploratory analysis, we observed that searching for values of a on an exponential scale allowed for more gradual control over the curve shape, and therefore we searched directly for values of e^a (effectively replacing a by e^a in Equation 1). We performed a grid search of 10,000 evenly spaced values between -5 and 15 (limits chosen to go beyond the fitted values), and selected the value of a yielding the curve minimizing the sum of squared errors to the values from the simulation.

To compare the efficiency curves within a given simulation (i.e., between samplers and optimization layers for a given metaweb and set of species ranges), we used the definite integral of Equation 1 between 0 and k , the maximum number of sampling sites in the landscape (10,000), which takes the form:

$$\int_0^k \frac{x}{a + x} dx = a \ln(a) - a \ln(k + a) + k \quad (2)$$

Using @eq-eff-int, we can compute the difference in efficiency between two curves as the difference of integrals:

$$d_{1,2} = (a_1 \ln(a_1) - a_1 \ln(k + a_1) + k) - (a_2 \ln(a_2) - a_2 \ln(k + a_2) + k) \quad (3)$$

This difference measure allows to identify which samplers or optimization layer is more efficient, quickly reaching a high proportion of monitored interactions (positive values indicate that the first curve is more efficient than the second), while also quantifying the magnitude of this difference in a comparable way between possible comparison pairs. We compared all samplers and optimization layers individually within each independent simulation, given that the monitoring context is different in each one (different metawebs and species ranges).

Results

Lower sampling expectations for species interactions than species richness

Our results for metaweb sampling highlight a marked difference between the proportions of sampled species and interactions (Figure 1). Sampled species saturated very quickly, with all species sampled with less than 25 sites in a BON across the landscape. Possible interactions, which depend on co-occurrence of species with a feasible interaction in the metaweb, also saturated quickly, reaching over 90% of interactions in the metaweb with 25 sites and increasing asymptotically afterwards. In contrast, realized and detected interactions, which account for the impact of species abundances, reach much lower proportions, both under 25% of the metaweb after 100 sites. While the proportion would keep increasing with a higher number of sites, the speed at which interactions accumulate is notably lower than for species richness and possible interactions, highlighting the need for a much higher expected sampling effort.

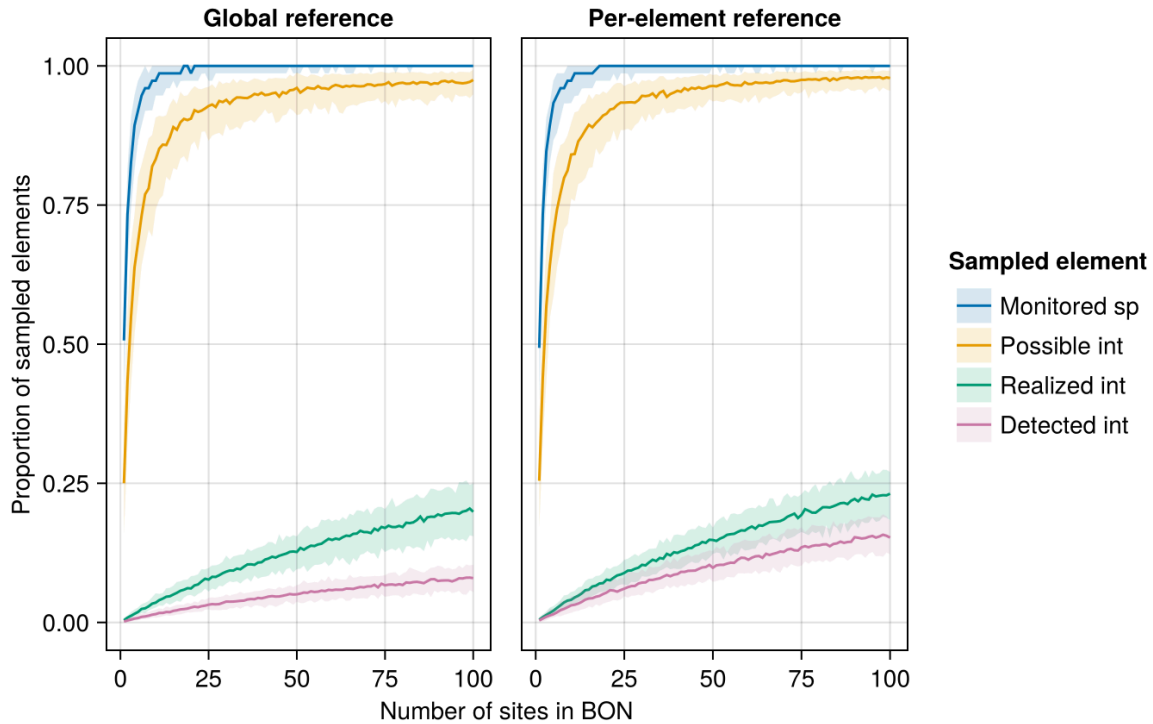


Figure 1: Proportion of sampled species and interactions for a given number of sampling sites in a Biodiversity Observation Network (BON) across the simulated landscape. The central line indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates. The global reference (left) uses total species richness and the total number of interactions in the metaweb as the maximum reference to calculate the proportion while the per-element reference (right) uses the maximum number of an interaction actually present in the simulated landscape.

Differences between samplers and optimization layers

Our focal sampling experiment showed the Uncertainty Sampling algorithm to be most efficient at optimizing BON-design based on a known species range and while aiming to maximize the sampling of interactions of a single focal species (Figure 2). Simple Random sampling, as a proxy for uninformed sampling, performed the worst among the algorithms tested. Weighted Balanced Acceptance performed notably less than Uncertainty Sampling at maximizing the efficiency for the focal species, although it maintained a better spatial balance across the landscape.

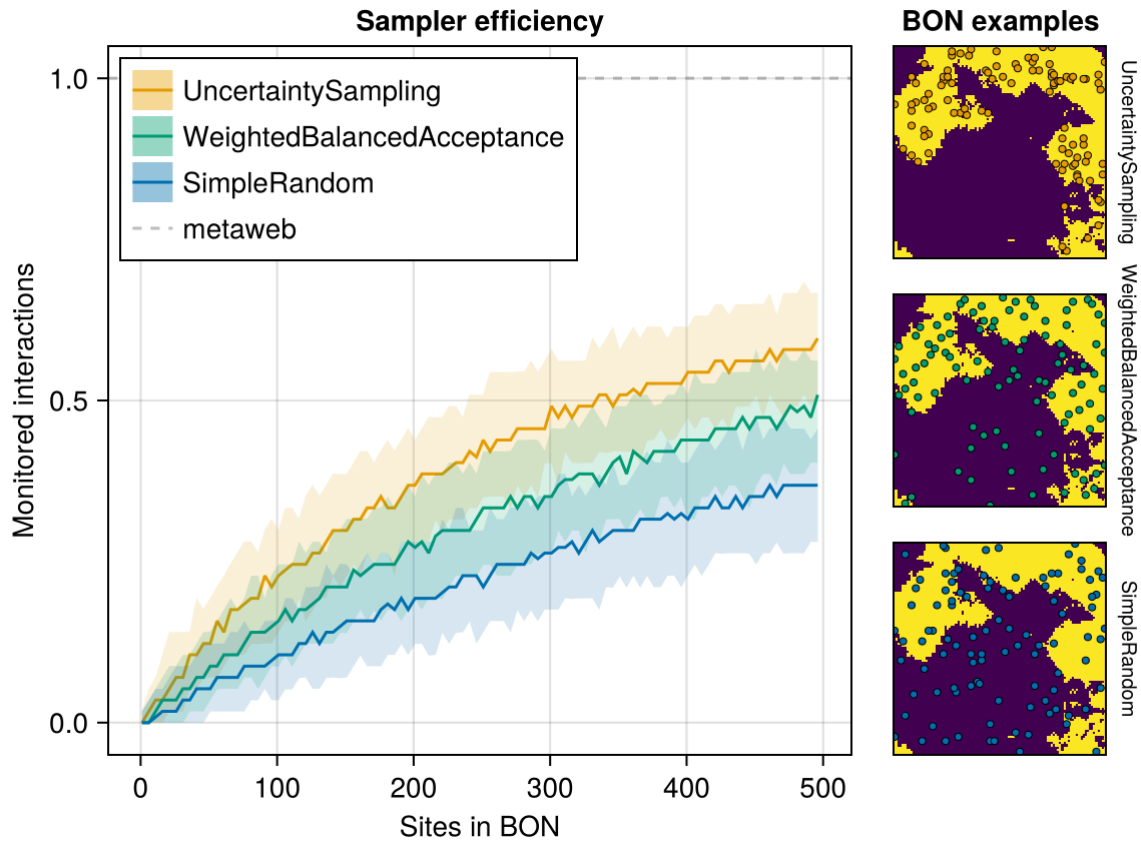


Figure 2: Efficiency of three sampler algorithms at sampling realized interactions for a given focal species within Biodiversity Observation Networks (BONs). The central line indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates. The right panels illustrate examples of BONs of 50 sites (coloured points) generated by each sampler over the focal species range (present at pixels in yellow).

Optimizing on the known species range, our proxy for an attainable level of information an interested user might have, was a more efficient strategy to retrieve the focal species' interactions than optimization based on species richness (or interaction richness), effectively sampling more than half of the species' interactions (Figure 3). In contrast, optimization based on the location of realized interactions, our best-case scenario requiring a higher level information, expectedly performed better, retrieving most of the species' interactions.

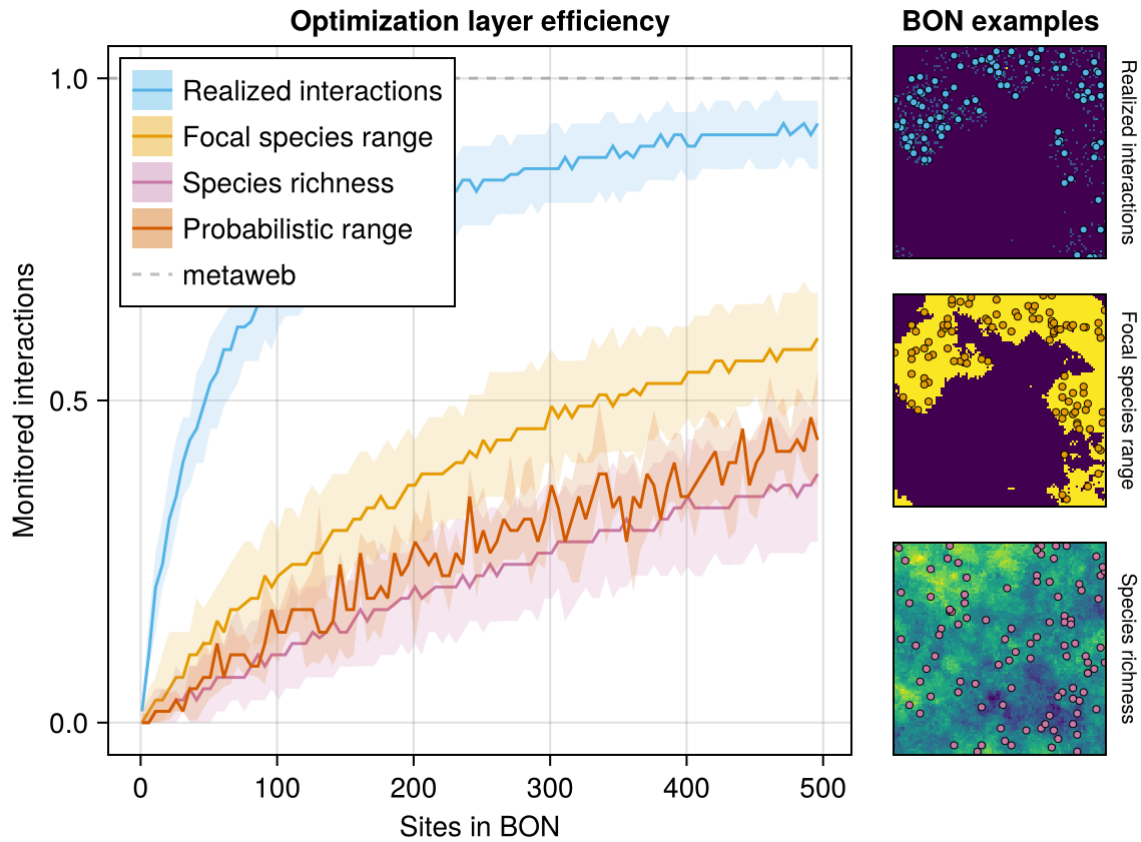
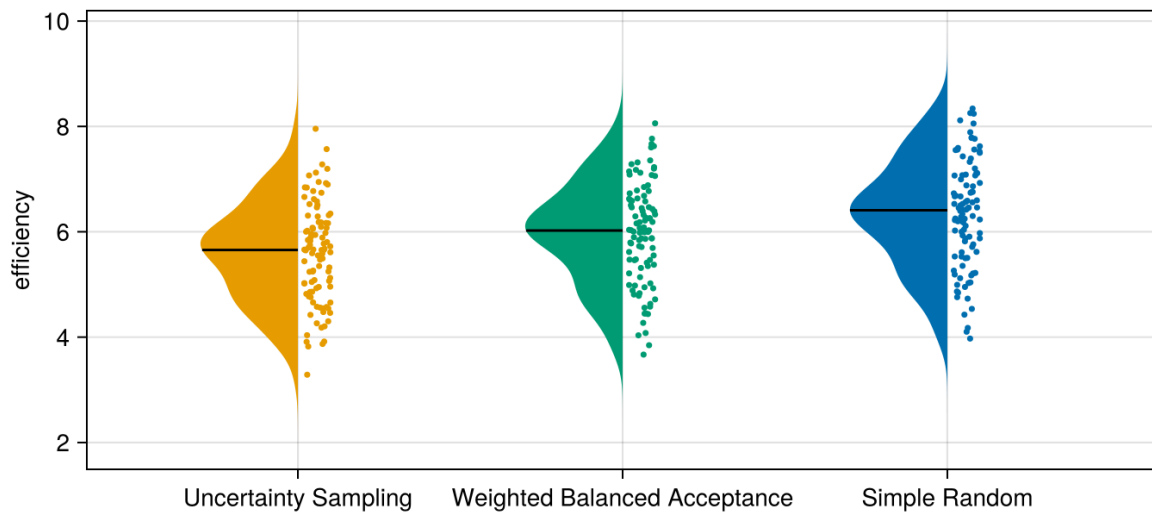


Figure 3: Efficiency of three optimization layers at sampling realized interactions for a given focal species within Biodiversity Observation Networks (BONs). The central line indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates. The right panels illustrate examples of BONs of 50 sites (coloured points) generated by each sampler over the focal species range (present at pixels in yellow).

Comparison of efficiency parameters across simulations

The distribution of efficiency parameters across independent simulations confirmed the interpretations based on Figure 2, Figure 3, but also highlighted a wide range of outcomes in efficiency between simulations. Among samplers, the median efficiency parameter indicated a better performance (lower parameter value) for Uncertainty Sampling, followed by Weighted Balanced Acceptance and Simple Random (Figure 4A). The actual efficiency values spanned a similar range across the possible sampler options with all simulations combined. Among optimization layers, optimizing on the focal species range performed better than on the probabilistic range, followed by species richness, although all three options spanned similar ranges with all simulations combined (Figure 4B). In contrast, optimizing based on the realized interactions distinctly and consistently performed better across all simulations.

A) Samplers



B) Optimization Layers

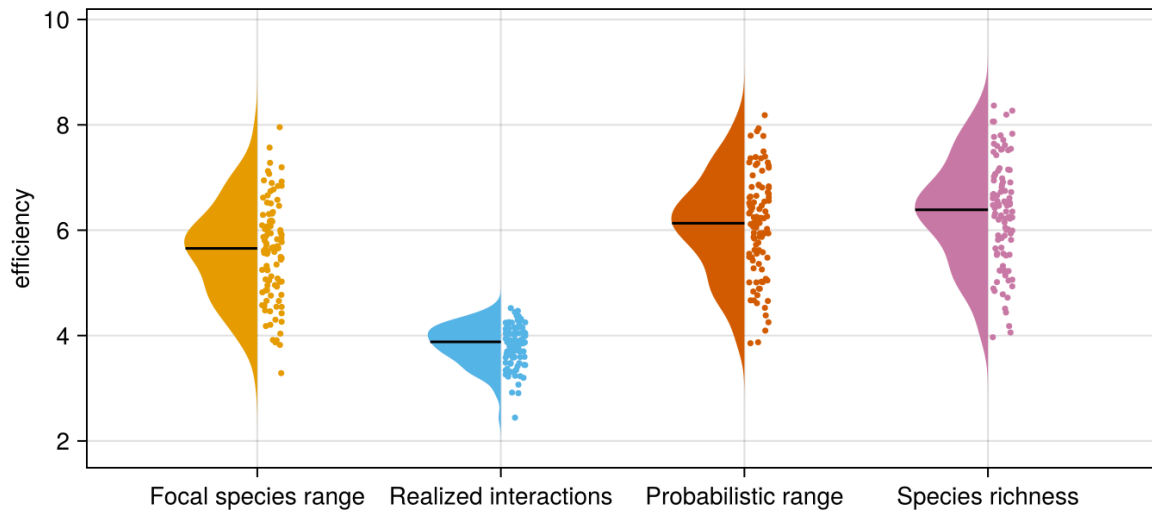
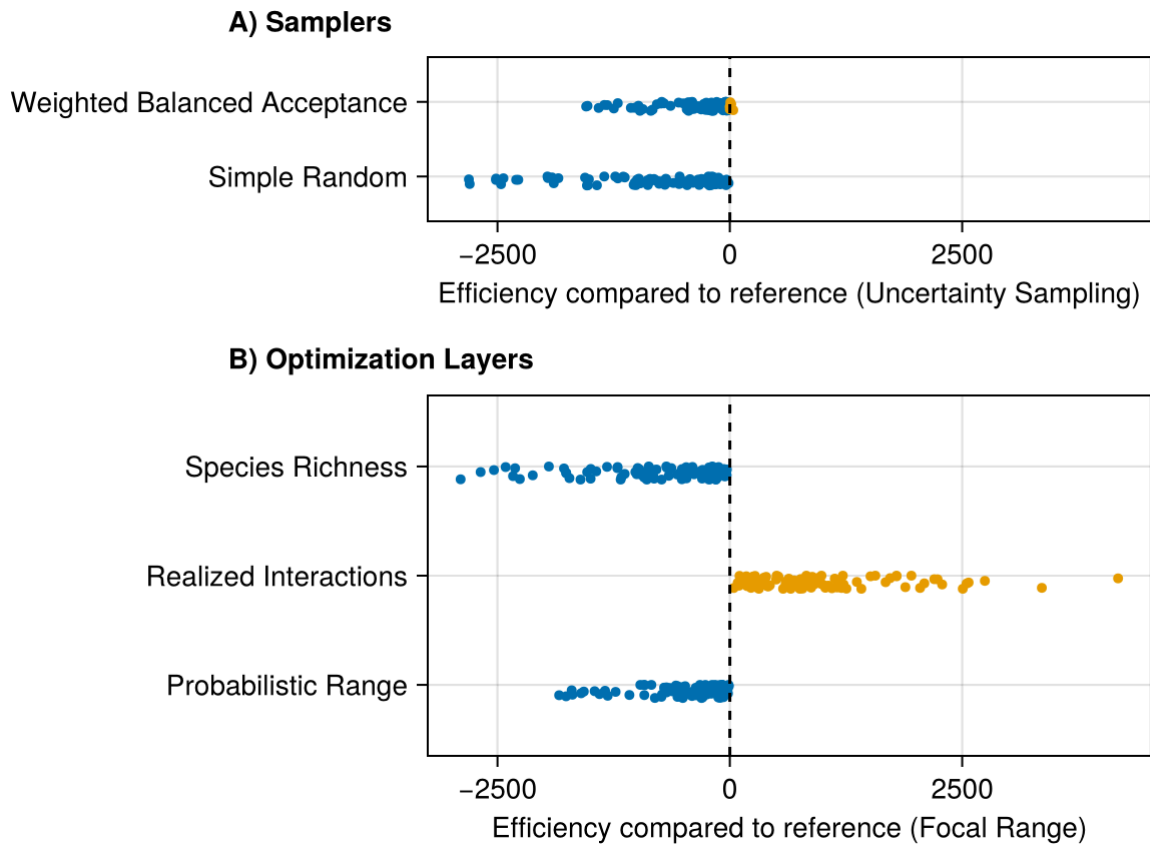


Figure 4: Distribution and density of efficiency parameters across 100 independent simulations for the three sampler options tested on the focal species range (A) and for the four optimization layer options using Uncertainty Sampling as the sampler (B). Uncertainty Sampling on panel A and Focal species range on panel B therefore represent the same data.

The fact that multiple options span similar ranges of efficiency values across independent simulations highlights the need to compare the efficiency within a given simulation. Comparing options one-by-one clearly highlighted a most efficient one in all case (Figure 5, Supp. Mat.), consistent with the showcased single-simulation example (Figure 2, Figure 3). Uncertainty Sampling was more efficient than Weighted Balanced Acceptance in 98 of 100 simulations, and both were more efficient than Simple Random in all simulations (Figure 5). Optimizing on the focal species range was always more efficient than on the probabilistic

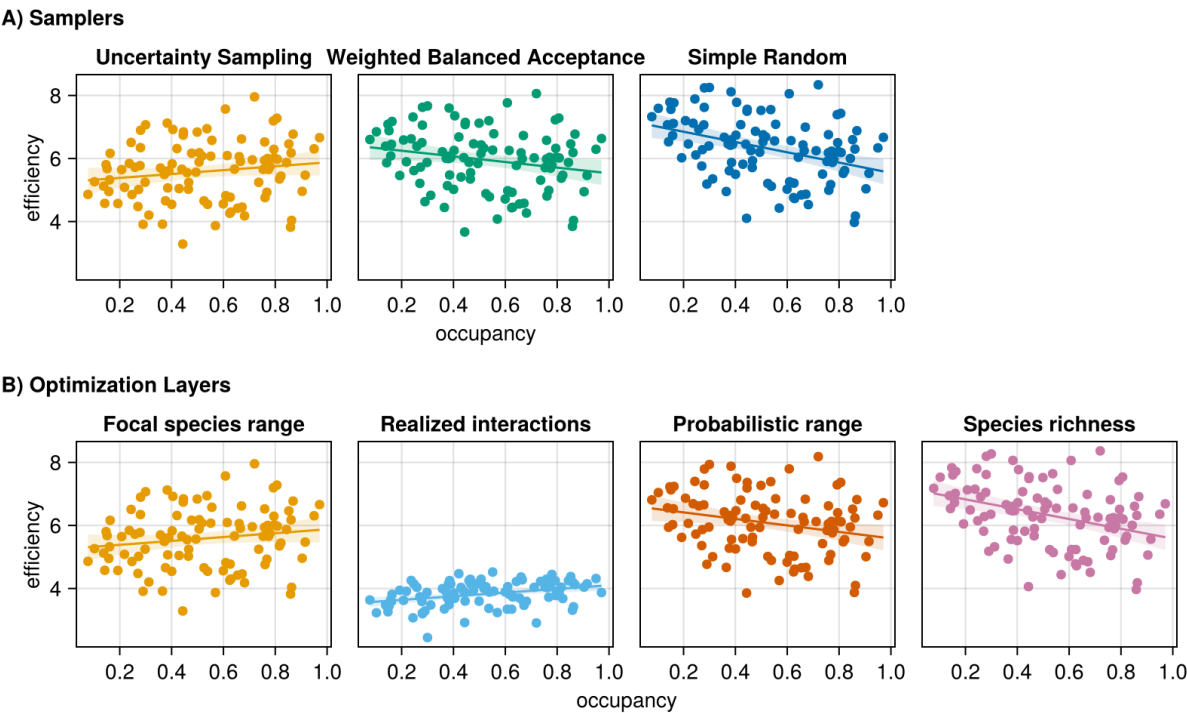
200 range or species richness (with probabilistic range more efficient than species richness in all
 201 but one simulation), but always less efficient optimizing on realized interactions.



202
 203 Figure 5: Within-simulation comparison of efficiency between samplers and optimization layers. The
 204 comparison values is based on the difference of the definite integrals of the efficiency samplers curves
 205 described in Equation 3. Negative values (in blue) indicate that the compared option lead to a less efficient
 206 sampling than its reference, while positive values (in orange) indicate it was more efficient. References were
 207 chosen for narrative purposes and to simplify the number of comparisons displayed. All one-by-one
 208 comparisons are available in Supp. Mat.

209 While the simulations clearly highlight which sampler or optimization layer is most efficient
 210 across all simulations, efficiency parameters and differences vary over a wide range of
 211 values. We verified if this could be tied to the occupancy of the focal species in simulated
 212 landscape, measured as the number of cells where the species is present. We examined the
 213 relationship between occupancy and efficiency. Optimizing using Uncertainty Sampling on
 214 the focal species range and on realized interactions is especially efficient at low occupancy, as
 215 it targets the fewer cells where the focal species is present. As species occupancy increases,
 216 the difference between options decreases. When the focal species has high occupancy and
 217 is present almost everywhere across the landscape, Weighted Balanced Acceptance and
 218 Simple Random perform similarly to Uncertainty Sampling, as balancing sites across space

219 or choosing at random becomes equivalent to targeting the species range. Optimizing based
 220 on species richness became more efficient as occupancy increased. A possible explanation is
 221 that at high occupancy, species range is not a restricting element and thus targeting sites with
 222 high richness directly leads to targeting sites with more interaction partners. On the contrary,
 223 at low occupancy, species range is an important restriction, and optimizing on richness of all
 224 species might target sites where the focal species is absent. Optimizing on the probabilistic
 225 species range behave similarly to species richness; at high occupancy, it is very likely the
 226 species will be present at the sites with high probabilities and therefore the sampler can more
 227 effectively filter out sites where the species is absent. At low occupancy, the focal species will
 228 be present in fewer locations with high probabilities, but might be absent from locations with
 229 intermediate probabilities, making it more challenging for the sampler to pick apart.



230
 231 Figure 6: Relationship between the occupancy of the focal species and sampling efficiency for all optimization
 232 options. These results come from 100 independent simulations, and therefore 100 different sets of metawebs,
 233 species ranges, and occupancy of the focal species.

234 Discussion

235 What should be realistic expectations for sampling and monitoring of species interactions?

- 236
- Not as high as species richness and not as simple as co-occurrence of interacting species

237 • Expectations should be much lower given abundances, and our simulations highlight a
 238 way this could happen

239 • This means we can apply motoring to species interactions, with limits.

240 We can improve coverage with the right optimization method. But to do so, we need a clear
 241 and attainable sampling or monitoring objective.

242 • Again, we come back to a single species perspective as most attainable and understand-
 243 able level of information while yielding relevant results.

244 • Optimizing on species range is an efficient strategy to retrieve its interactions (most
 245 attainable information), better than optimization based on species or interaction richness
 246 and surpassed only by knowledge of it's realized interactions

247 Our results offer insights into defining realistic expectations for interaction monitoring. With
 248 the right target and optimization strategy, available information, such as species ranges, can
 249 guide observation network designs to monitor species interactions with a reasonable number
 250 of sites.

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